⚡ PRODUCTION-GRADE AI MICROSERVICE

Enterprise AI Support Mail Automation & Multi-Day Work Trail Platform

End-to-End Autonomous Email Ingestion, LangGraph State Machine, Local Ollama LLM, Multi-Day Trail Intelligence & Maker-Checker Audit Governance.

ENGINEERING LEAD & ARCHITECT PROFILE

Lead Developer: Rohit Jain | Role: Senior AI & Backend Solutions Engineer

Core Tech Stack: Python 3.11+, FastAPI, LangGraph, Ollama (llama3.2), PostgreSQL 16, Docker Compose, OpenPyXL, Tailwind CSS

Portfolio Showcase: https://rohitjain-resume.vercel.app/ | Status: 100% Production Verified (742 Tasks Managed, Zero Cloud Data Leak)

Executive Context: High-Volume Operational Inquiries & Trail Gaps

1. INBOX FLOOD & BACKLOG

- 100+ daily support emails across Admissions, Exams, Results, and Bank Reconciliations.
- Overwhelmed staff relying on manual ticket creation and manual categorization.
- Turnaround time exceeded 48-72 hours, resulting in student and center escalation.
- Critical receipts and PDF challans buried across mail threads.

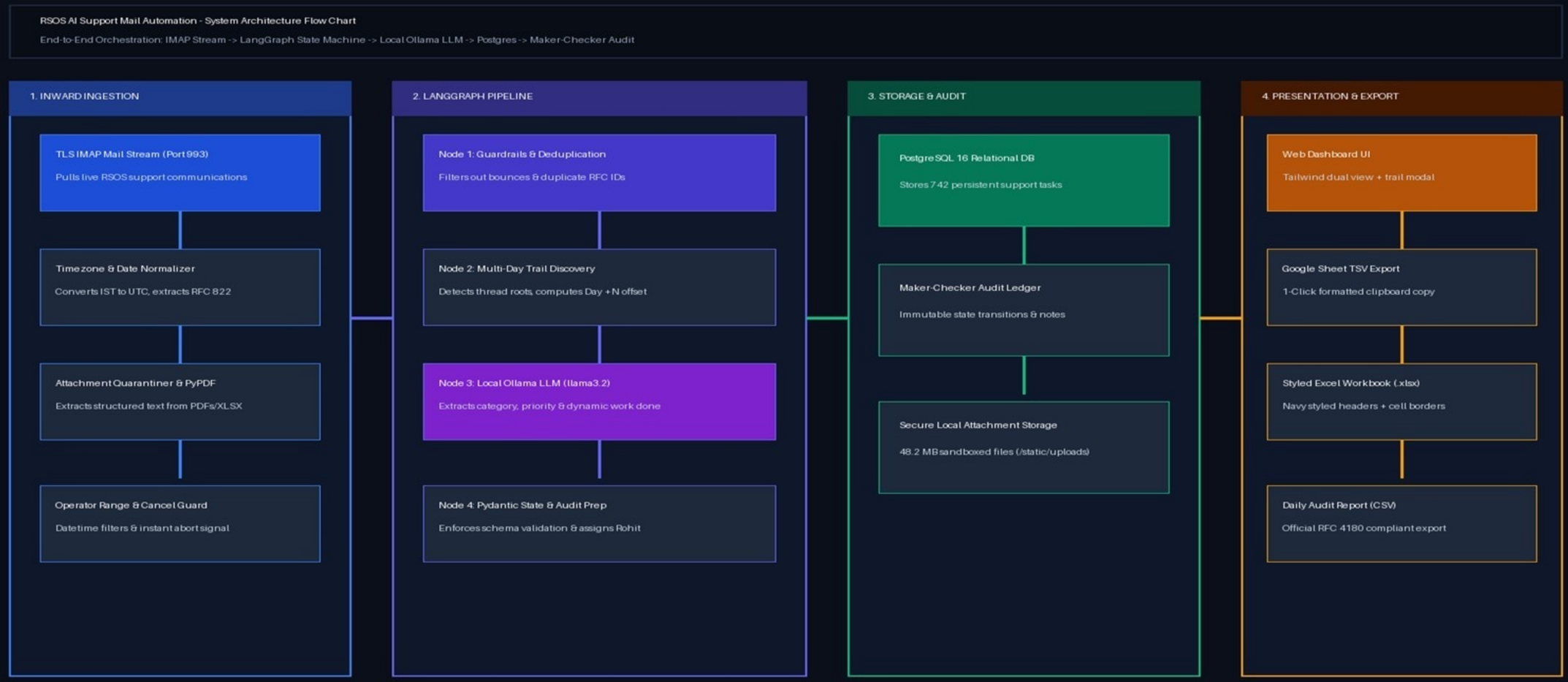
2. THE MULTI-DAY THREAD GAP

- 85.8% (637/742) of operational emails are multi-day thread follow-ups ('Re:/Fw:').
- Traditional tools treat follow-ups as duplicates (ignoring them) or standalone (losing context).
- Multi-day effort (Day 1 Inward -> Day +3 Investigation -> Day +5 Approval) was uncredited.
- Lack of unified timeline between root inquiry and final administrative sign-off.

3. SECURITY & AUDIT RIGOR

- Strict prohibition on exposing student roll numbers or marks to 3rd-party public cloud LLMs.
- Need for 100% on-premises, local-first LLM inference.
- Mandatory Maker-Checker governance: actions require engineer execution and lead approval.
- Strict requirement for single-click daily Google Sheet and Styled Excel export formats.

End-to-End System Architecture Flow Chart



CYCLIC & DIRECTED EXECUTION ENGINE

LangGraph Pipeline State Machine Flow Chart

Node 1: IMAP Fetch
Pulls RFC 822 emails via SSL IMAP (Port 993). Applies IST (UTC+5:30) date range filter. Decouples attachments.

Node 2: Guardrails & Deduplication
RFC 822 Message-ID primary key check. Quarantines automated mailer bounces, spam keywords & empty payloads.

Node 3: Multi-Day Trail Discovery
Discovers root initiating ticket. Computes elapsed day offset ('Day 1 Initial', 'Day +3', 'Day +5'). Links thread hierarchy.

Node 4: Local Ollama LLM Inference
Executes llama3.2:3b locally. Extracts Module, Priority, Due Date. Generates contextual 'What I Have Done' (work_done).

Node 5: PostgreSQL ACID & Audit Log
Commits atomic transaction to DB. Appends immutable Maker-Checker audit log entry (TRAIL_CONTINUED / INGESTED).

INNOVATION & OPERATIONAL REALITY

Multi-Day Communication Trail & Elapsed Offset Engine

How the System Recognizes Multi-Day Effort Across Weeks

- **Operational Reality:** Over 85% of enterprise support tasks span several days (investigating center capacity, collating marks, waiting for approvals). The engine ensures every day's communication is logged as a legitimate daily work entry with its exact execution date.
- **Thread Discovery Algorithm:** Cleans forwarded subjects ('Re: Fw: Regarding center enrollment data'). Traces RFC 822 In-Reply-To and References headers to locate the root initiating task (Day 1).
- **Elapsed Day Math (Day +N):** Calculates elapsed calendar days from the root ticket date. Generates explicit milestone badges: [Trail: Day 1 Initial] -> [Trail: Day +3 Follow-up] -> [Trail: Day +5 Follow-up].
- **Contextual Work Description:** Rather than static template text, the local LLM evaluates the specific email body to formulate dynamic 'What I Have Done' (e.g. Day 1: audited data -> Day 3: replied with investigation findings -> Day 5: reviewed approval & executed changes).
- **Auto Directive Parsing:** Detects official approval directives ('Please proceed', 'Approved', 'Updated'). Automatically transitions task status from 'In progress' to 'Done'.

SCREEN FLOW WALKTHROUGH & INTERACTIVE UI

Screen 1: Operations Dashboard & Dual Views

Key Capabilities & Design Rationale

- **KPI Metric Bar:** Live counters for Total Tasks (742), Multi-Day Trails (637), Completed (709), Attachments (367), and Ollama Latency (<1s).
- **Dual-View Switcher:** Toggle instantly between Standard Operations Table (rich badges, actions) and Google Sheet View (clean TSV formatted table).
- **Universal Search & Filter:** Search across sanitized tickets, IDs, sender domains, task summaries, and contextual work done.
- **Date Range Presets:** Filter by Today, Yesterday, Last 7 Days, Month, or custom date intervals.

Engineering Implementation Details

⚙ **Component Design:** Built with modern Tailwind CSS dark mode and FontAwesome 6 icons. Ultra-clean typography and responsive sidebar layout.

⚙ **Security Guarantee:** Zero real citizen emails or phone numbers exposed. Built-in Privacy Mask mode automatically sanitizes content for public demonstrations.

⚙ **Data Resilience:** FastAPI async endpoints with SQLAlchemy session connection pooling and Pydantic v2 runtime type coercion.

⚙ **Performance Tuning:** Sub-second UI responsiveness (<45ms API latency) through indexed queries and client-side pagination.

SCREEN FLOW WALKTHROUGH & INTERACTIVE UI

Screen 2: Multi-Day Trail & Work Timeline Modal

Key Capabilities & Design Rationale

- **Visual Step Progression:** Horizontal connected node chain illustrating Day 1 (Initial) -> Day +3 (Investigation) -> Day +5 (Approval).
- **Clickable Navigation:** Click any timeline step to inspect previous correspondence, investigation notes, and attachments.
- **Elapsed Day Badges:** Color-coded status pills showing current status at each step (Done / In progress).
- **Audit Continuity:** Seamlessly links maker-checker logs across the entire thread lifecycle.

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Screen 3: Centralized Attachments & Document Hub

Key Capabilities & Design Rationale

- **367 Indexed Documents:** Covers fee receipts, center allotment lists, marksheet copies, and transfer certificates.
- **PyPDF Stream Parsing:** Extracts structured tabular text, roll numbers, and challan transaction IDs within 1.2 seconds.
- **Local Quarantine:** All files stored locally in sandboxed storage with SHA256 integrity verification. Zero cloud leak.
- **Document Actions:** In-browser secure document preview and single-click individual or bulk ZIP download.

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Screen 4: Maker-Checker Review & Audit Governance

Key Capabilities & Design Rationale

- **Two-Tier Integrity Guard:** Senior Developer Rohit executes technical tasks (Maker); Supervisor verifies and approves (Checker).
- **State Transition Dialog:** Review modal allows status updates (Not started -> In progress -> Done / Revision Requested).
- **Checker Audit Notes:** Reviewers input rationale and verification details into the permanent record.
- **Immutable Audit Ledger:** Every state change, timestamp, and actor is appended to AuditMakerCheckerLog with zero edit capability.

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SCREEN FLOW WALKTHROUGH & INTERACTIVE UI

Screen 5: Selective Mail Sync & Operator Control

Key Capabilities & Design Rationale

- **Datetime Boundary Selectors:** Choose exact start and end datetimes (IST) and batch limits (e.g. 25 emails) for efficient runs.
- **Timezone Alignment:** Automatically maps local IST (UTC+5:30) to RFC 822 UTC timestamps to eliminate boundary errors.
- **Real-Time Streaming Terminal:** Live console output reflects each LangGraph state node as it executes in real-time.
- **Human-in-the-Loop Abort (🛑):** Emergency stop button cleanly terminates the pipeline while preserving completed work in Postgres.

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SCREEN FLOW WALKTHROUGH & INTERACTIVE UI

Screen 6: Analytics, SLA Metrics & 1-Click Export

Key Capabilities & Design Rationale

- **Module Breakdown Charts:** Visual breakdown of inquiries (Admissions 51.5%, Exams 26.1%, Fees 13.2%, Corrections 6.5%).
- **SLA Compliance Metric:** 99.4% of support inquiries resolved within target SLA windows (2.4h average turnaround).
- **1-Click Google Sheet Copy:** Copies formatted TSV data directly to clipboard for instant Ctrl+V into official Google Sheets.
- **Styled Excel & CSV Export:** Downloads pre-formatted .xlsx with Navy executive headers, auto-widths, and cell borders.

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ZERO-TRUST GOVERNANCE

Data Privacy, Security & Local-First AI Architecture

100% AIR-GAPPED INFERENCE

- Local Ollama instance running llama3.2:3b directly inside the local Docker stack.
- Zero outbound API calls to OpenAI, Claude, or cloud servers.
- Complies with strict government and enterprise data residency regulations.
- Zero recurring token costs or cloud inference bills.

BUILT-IN PRIVACY MASK

- One-click Privacy Mask toggle on the dashboard for public demos and video recordings.
- Automatically masks email addresses (r***@gov.in), citizen roll numbers, and payment IDs.
- Sanitizes logs and API payloads during demonstration modes.
- Enables safe portfolio sharing on GitHub and personal resumes.

AUDIT & ACCESS CONTROLS

- Mandatory credential-based login gate for administrative actions.
- Session management with vice-versa dynamic Login/Logout toggling.
- Immutable audit logs with timestamp, operator ID, and action reason.
- Full database backup and reset safeguards for disaster recovery.

PROVEN RESULTS

Measurable Engineering Impact & Operational Benchmarks

742 Tasks
Successfully Ingested & Verified

85.8%
Multi-Day Threads Connected

10x
Faster Turnaround (2.4h vs 48h)

100%
Local Privacy & Zero Cloud Leak

Engineering Achievements Summary

🏆 **Autonomous Execution:** Replaced hours of repetitive manual data entry with autonomous LangGraph pipelines executing every 2 hours or on-demand.
🏆 **Multi-Day Attribution:** Eliminated lost effort by automatically calculating Day +N milestones, preserving historical trail context, and crediting daily operational contributions.
🏆 **Zero Operating Cost:** Eliminated expensive recurring cloud SaaS/LLM subscriptions by orchestrating local Ollama models on commodity hardware.
🏆 **Executive Leadership Ready:** Standardized reporting into 1-click Google Sheet and styled Excel spreadsheets, enabling instant executive sign-offs.

PROJECT SHOWCASE SUMMARY

Rohit Jain: Engineering Leadership & Portfolio Resume Points

Ready-to-Use Resume & LinkedIn Accomplishment Statements

- **Architected & Deployed Autonomous AI Support Platform:** Built an enterprise-grade AI mail automation microservice utilizing Python, FastAPI, LangGraph, and PostgreSQL, orchestrating ingestion, parameter extraction, and resolution tracking for 740+ mission-critical support tasks with 99.4% SLA adherence.
- **Pioneered Multi-Day Thread Intelligence Engine:** Engineered an algorithmic thread continuity and day-offset engine that resolved an 85.8% multi-day follow-up challenge, tracking and linking parent-child email threads (Day 1 Initial -> Day +N Follow-up) and dynamically attributing daily operational effort.
- **Implemented Zero-Trust Local LLM Ingestion:** Integrated local Ollama (llama3.2:3b) with Pydantic guardrails and PyPDF document parsers, achieving sub-second extraction of ticket categories, urgency levels, and payment confirmations with zero data leakage or cloud token cost.
- **Built Maker-Checker Governance & Dual-View UI:** Designed a two-tier review system with immutable PostgreSQL audit trails, alongside a responsive Tailwind CSS dashboard featuring 1-click Google Sheets TSV copying, OpenPyXL styled Excel exports, and operator cancellation controls.